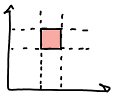
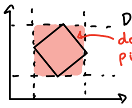
- Savi's code extracted patches using x_min, x_max, y_min, y_max
- but this no longer works bc we have patches that aren't parallel to the xy axis



Works here



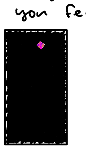
Doesn't work here

don't want these pixels captured

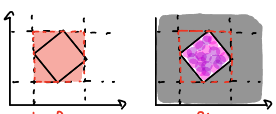
Step 1: use cv2.fillpoly() to take the pixels of the patch

- cv2.fillpoly takes a mask, any number of vertices, and a color as input
- the mask is just empty pixels the shape of the image
- the vertices are the 4 corners of the patch
- color = 255 (white)
- we are left with a mask of the patch

Step 2: Apply the mask to the tissue image using cv2.bitwise_and
- normally used to combine 2 given images
- if you feed it the same image twice and a mask it takes only that region of the image



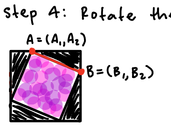
Step 3: Crop image around the patch
- uses same method as savi's original code
- isn't problematic anymore bc it will still capture the extra pixels but we blacked those out



before　　after

Result:

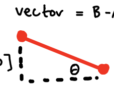
Step 4: Rotate the patch to be parallel to the axis



- need to find angle to rotate by

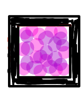
use this vector

vector = B - A = $(B_1 - A_1, B_2 - A_2)$



$\tan(\theta) = \frac{\text{vector}[0]}{\text{vector}[1]}$

$\theta = \tan^{-1}(\text{vector}[0] / \text{vector}[1])$

- rotate square by θ around the square's center point 

- finally, remove the extra pixels around the edges